

4.1 Related Rates

Motivation:

Suppose a conical tank is draining water slowly over time.



At each instant in time the volume of the water in the tank would be given by the equation

The volume of the water in the tank (V) would be changing over time (t).

If we were interested in finding the rate of change of the volume $\left(\frac{dV}{dt}\right)$ we could differentiate both sides of the equation above with respect to t .

So to find the value of $\left(\frac{dV}{dt}\right)$ at a specific point in time we need the values of r , h , $\frac{dr}{dt}$, $\frac{dh}{dt}$ at that specific time.

NOTE: The type of problem described above is called a *related rates problem*.

Steps to follow:

(1)

(2)

(3)

Note: A common error is for students to go to step 3 too early. Remember that you need to be working with variables when you take a derivative. It IS possible to plug in for variables that DO NOT change in step 1 (i.e. values that are fixed in the context of the problem).

Example 1: Assume that oil has spilled from a ruptured tanker and spreads in a circular pattern whose radius is increasing at a constant rate of 2 ft/s . How fast is the area of the spill increasing at the moment in time when the radius of the spill is 60 ft?

Example 2: Car A is traveling west at 50 mi/h and car B is traveling north at 60 mi/h . Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3mi and car B is 0.4mi from the intersection?

Example 3: An airplane is flying in a straight direction and at a constant height of 5 kilometers. Let the angle of elevation of the airplane from a fixed point of observation be called θ . The speed of the airplane is $500\text{km} / \text{h}$. What is the rate of change of angle θ when it is 45° ?

Example 4: A rectangular water tank is being filled at the constant rate of $0.02\text{m}^3 / \text{s}$. The base of the tank has dimensions $w=1\text{m}$ & $l=2\text{m}$. What is the rate of change of the height of water in the tank when the height of the water is 1m ?

Example 5:

Suppose that liquid is draining from a conical tank with a radius of 4cm and a height of 16cm . Suppose that the liquid drains at a constant rate of $2\text{cm}^3 / \text{min}$. At what rate is the depth of the liquid changing at the instant when the liquid in the cone is 8cm deep?